INCH-POUND
MIL-DTL-16377/27C(SH)
30 June 2014
SUPERSEDING
MIL-F-16377/27B(SH)
7 October 1988

DETAIL SPECIFICATION SHEET

FIXTURES, LIGHTING; INCANDESCENT AND SSL EQUIVALENT, GENERAL LIGHTING 50 WATTS, 120 VOLTS, WATERTIGHT, SYMBOLS 92.2, 93.2, 93.4, 93.5, 92.2SSL, 93.2SSL, 93.4SSL, AND 93.5SSL

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-16377.

All dimensions are in inches. For tolerances other than specified herein, see 3.5.2 of MIL-DTL-16377.

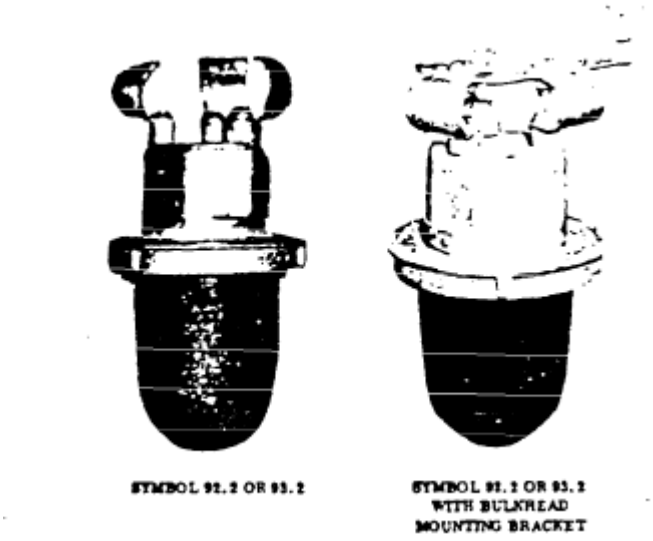


FIGURE 1. Fixtures, incandescent, symbols 92.2 and 93.2.

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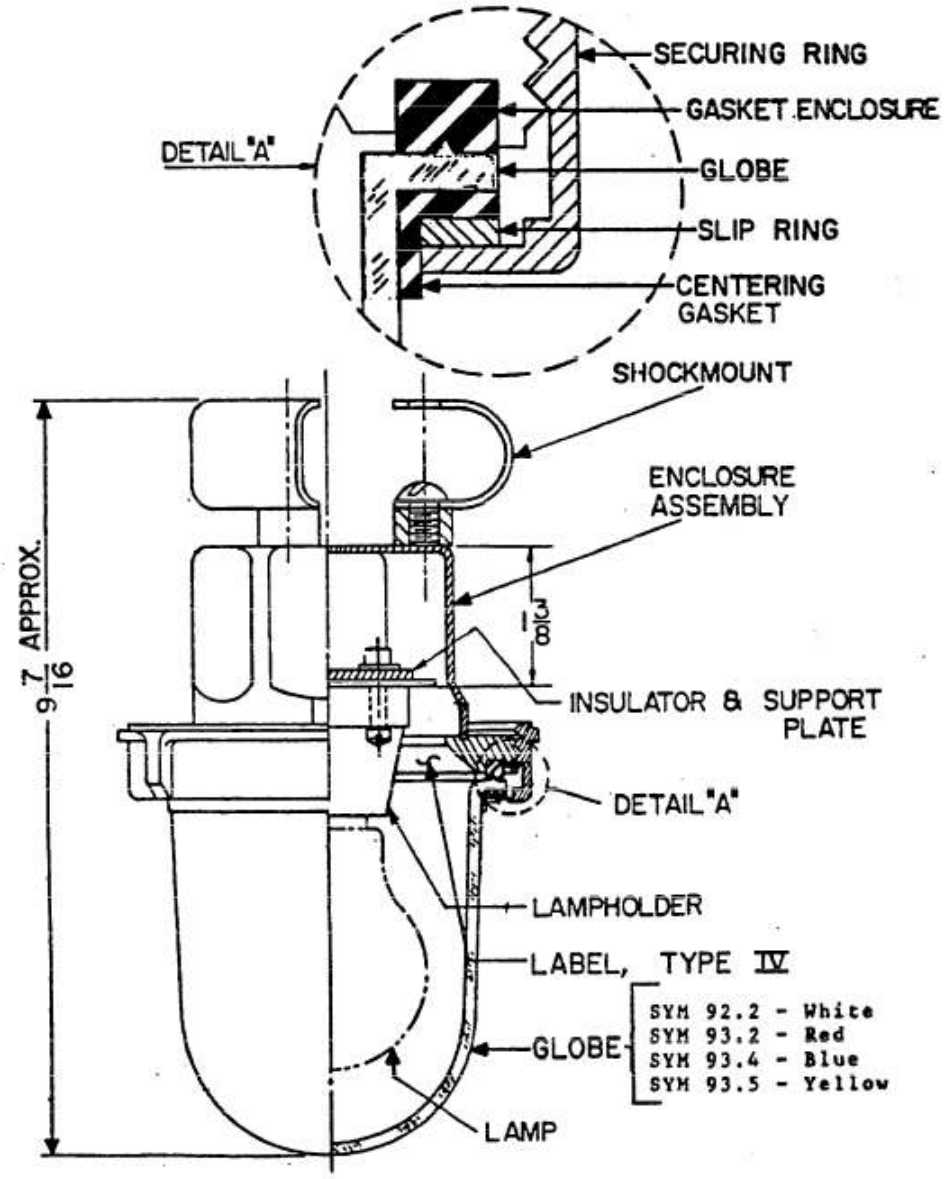


FIGURE 2. Fixture assembly.

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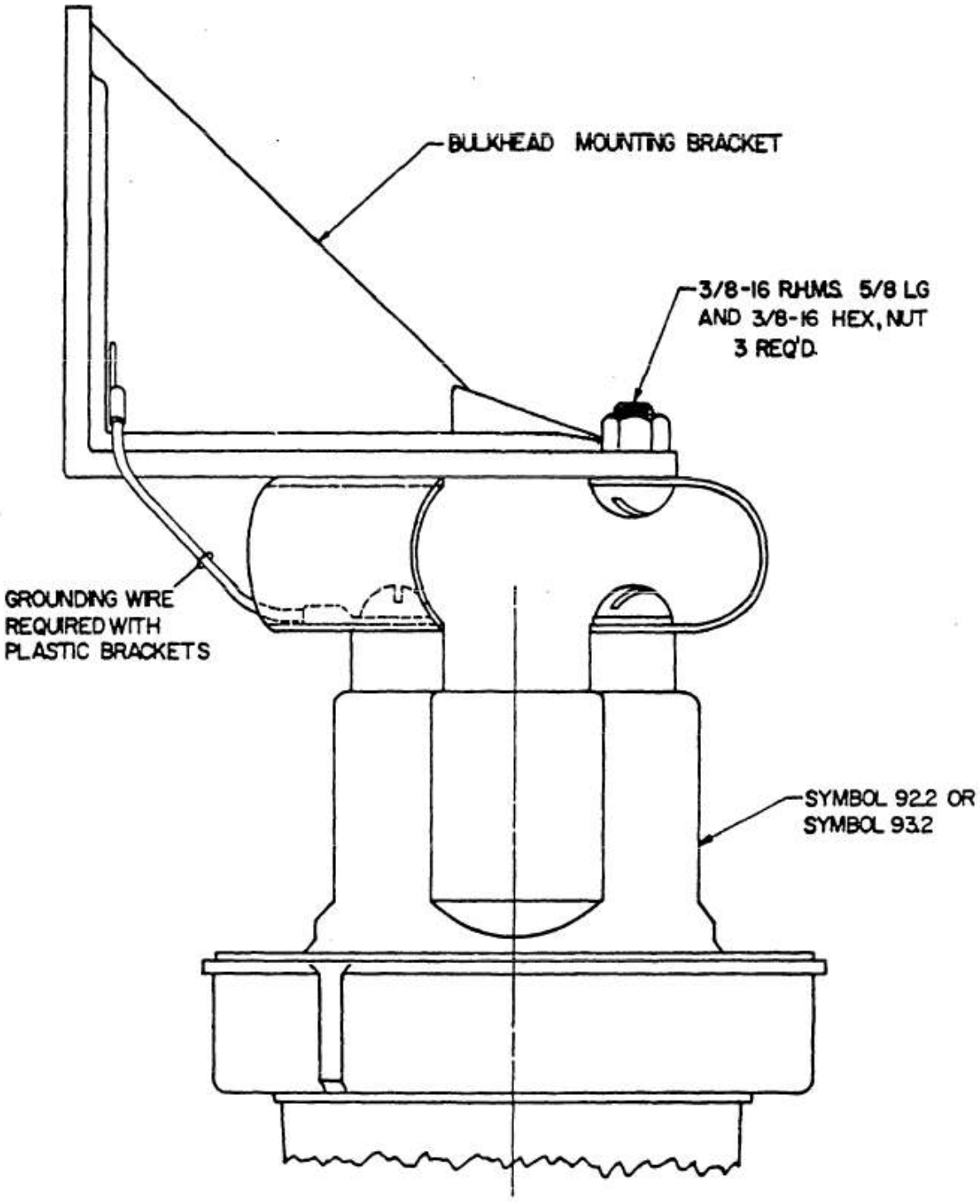


FIGURE 3. Fixture assembly for bulkhead mounting.

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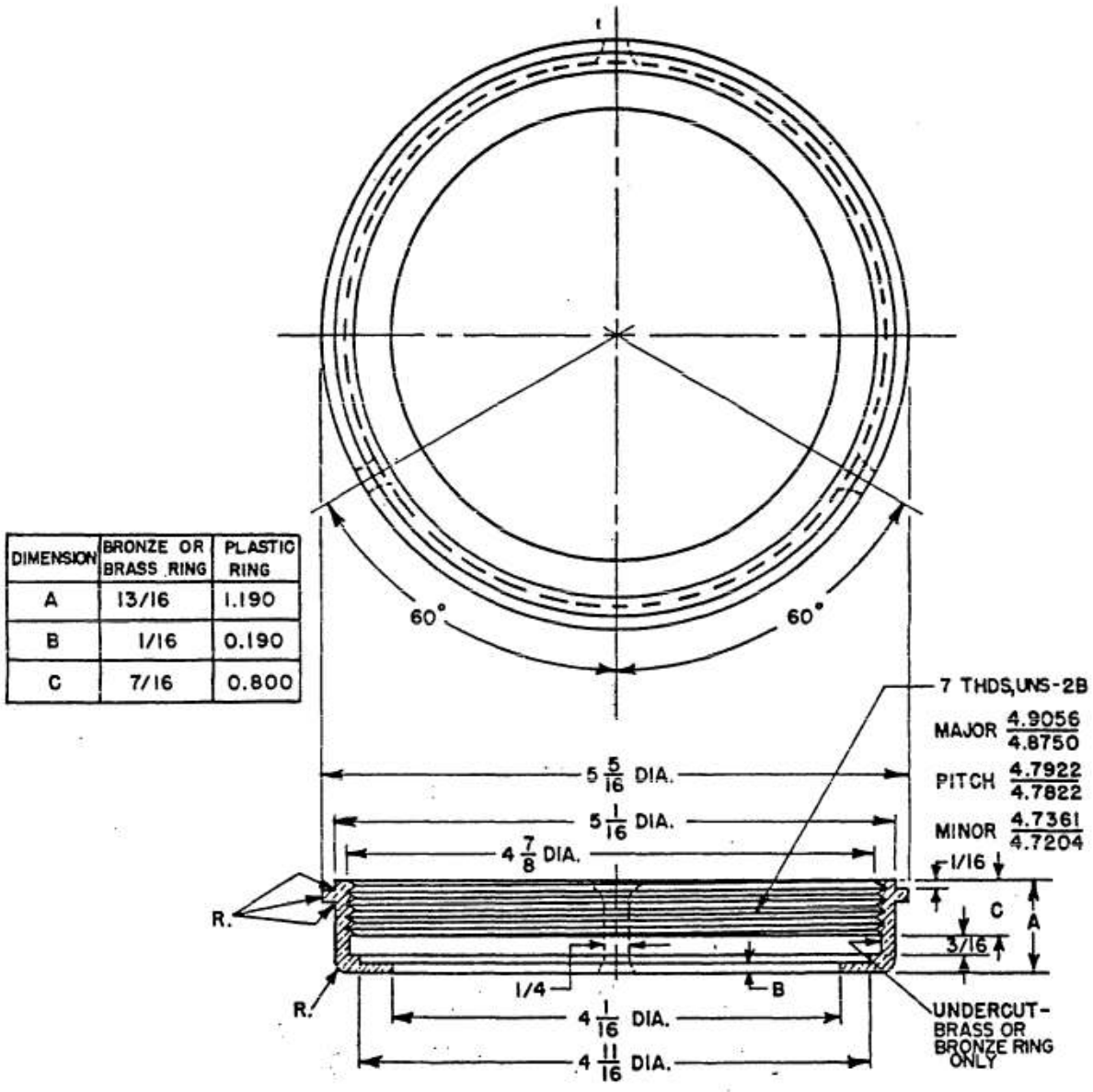


FIGURE 4. Securing Ring.

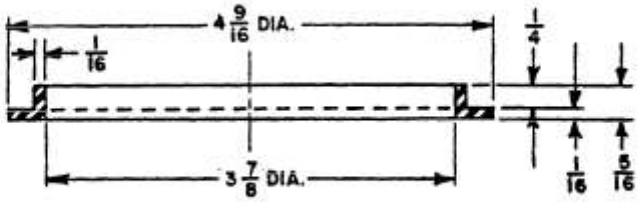


FIGURE 5. Globe centering gasket.

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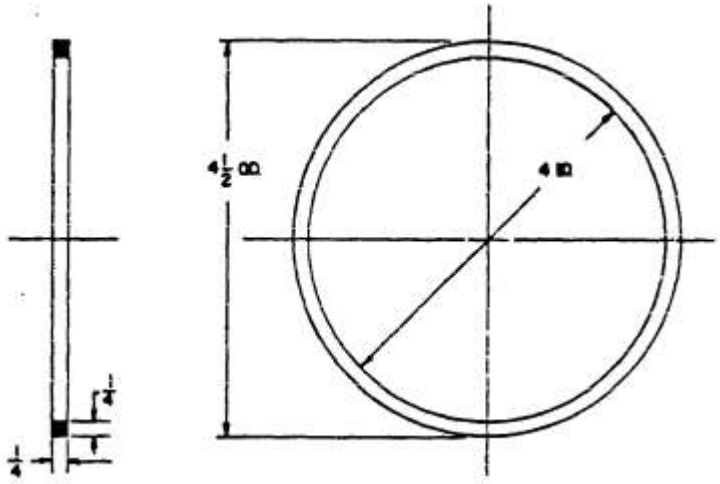


FIGURE 6. Enclosure gasket.

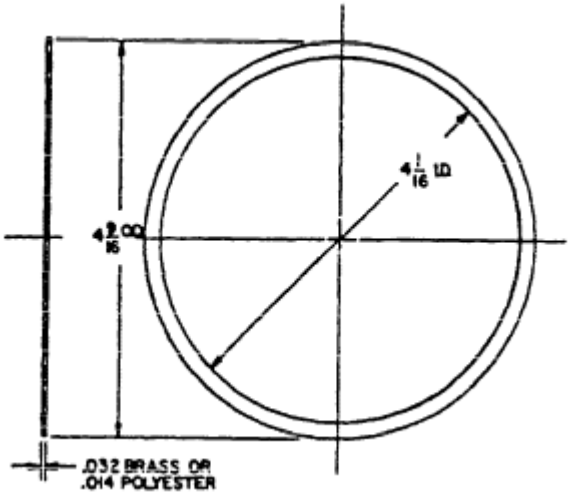


FIGURE 7. Slip ring.

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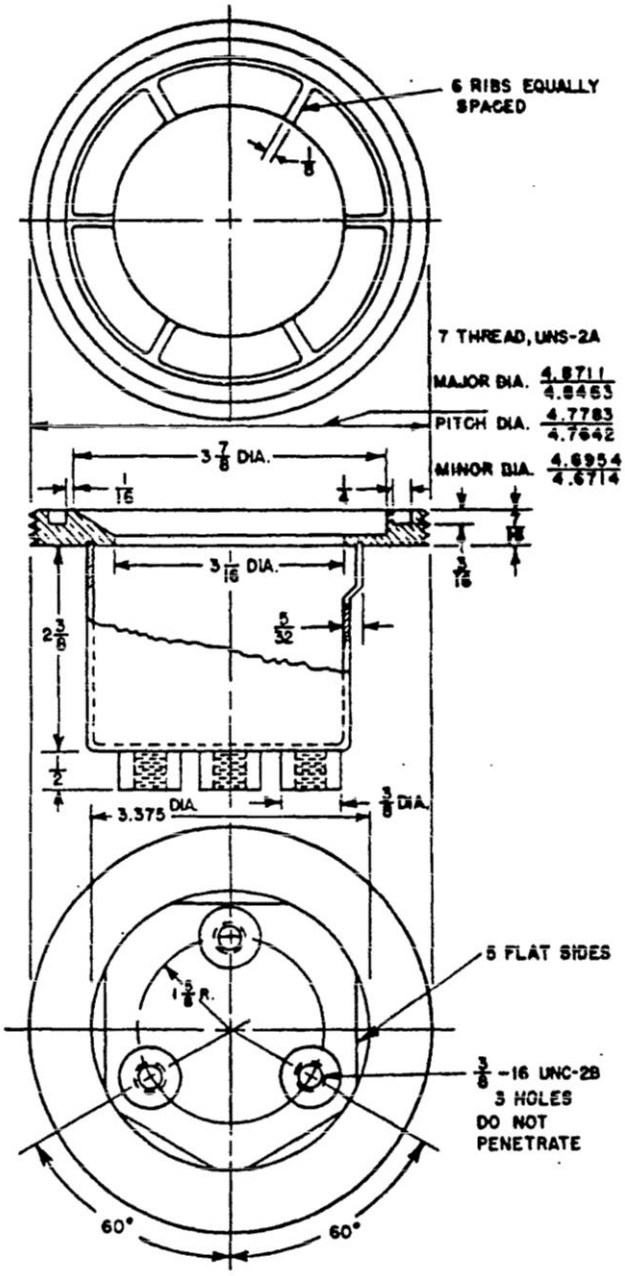
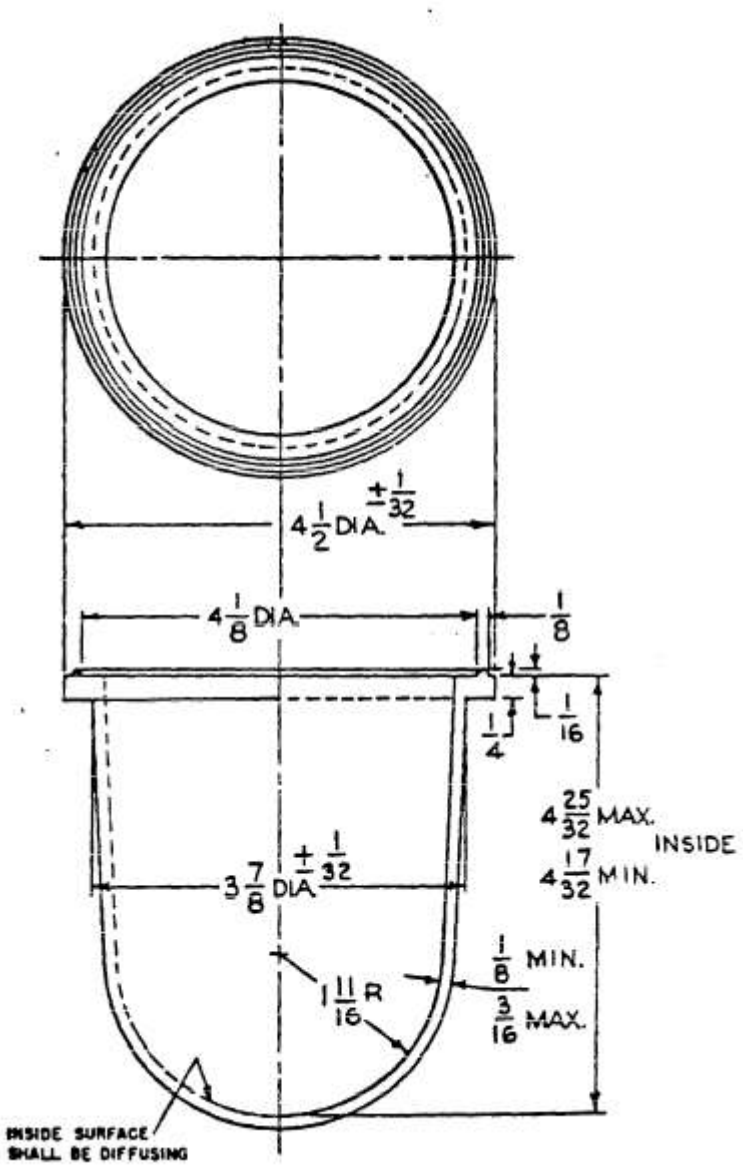


FIGURE 8. Enclosure assembly.

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Material: Tempered glass.
Color:

Symbol 92.2: Clear transparent/diffusing.
Symbol 93.2: Red transparent/diffusing.
Symbol 93.4: Blue transparent/diffusing.
Symbol 93.5: Yellow transparent/diffusing.

Transmittance (Symbol 92.2): Clear transparent/diffusing – 80 percent minimum.

FIGURE 9. Globe.

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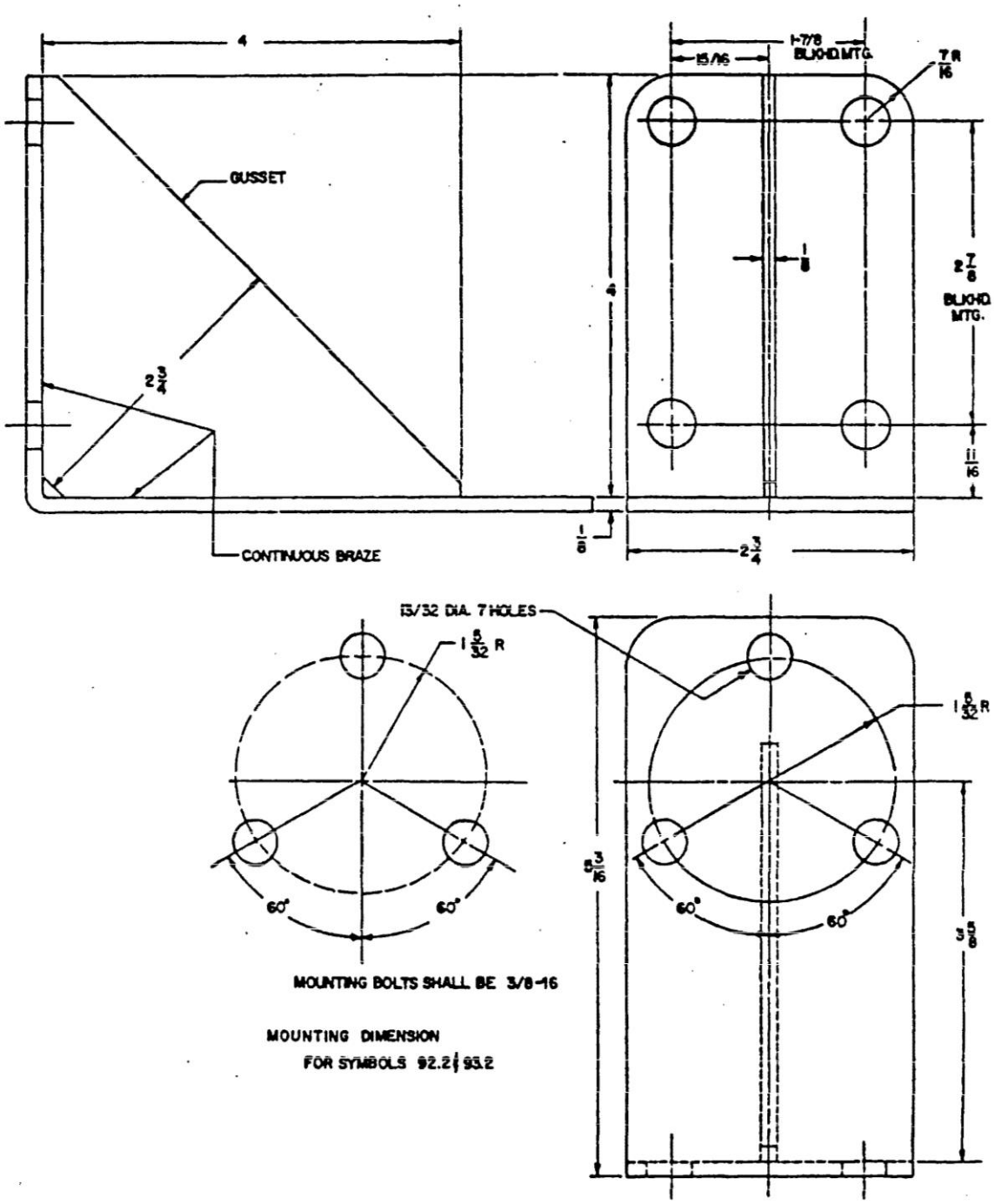
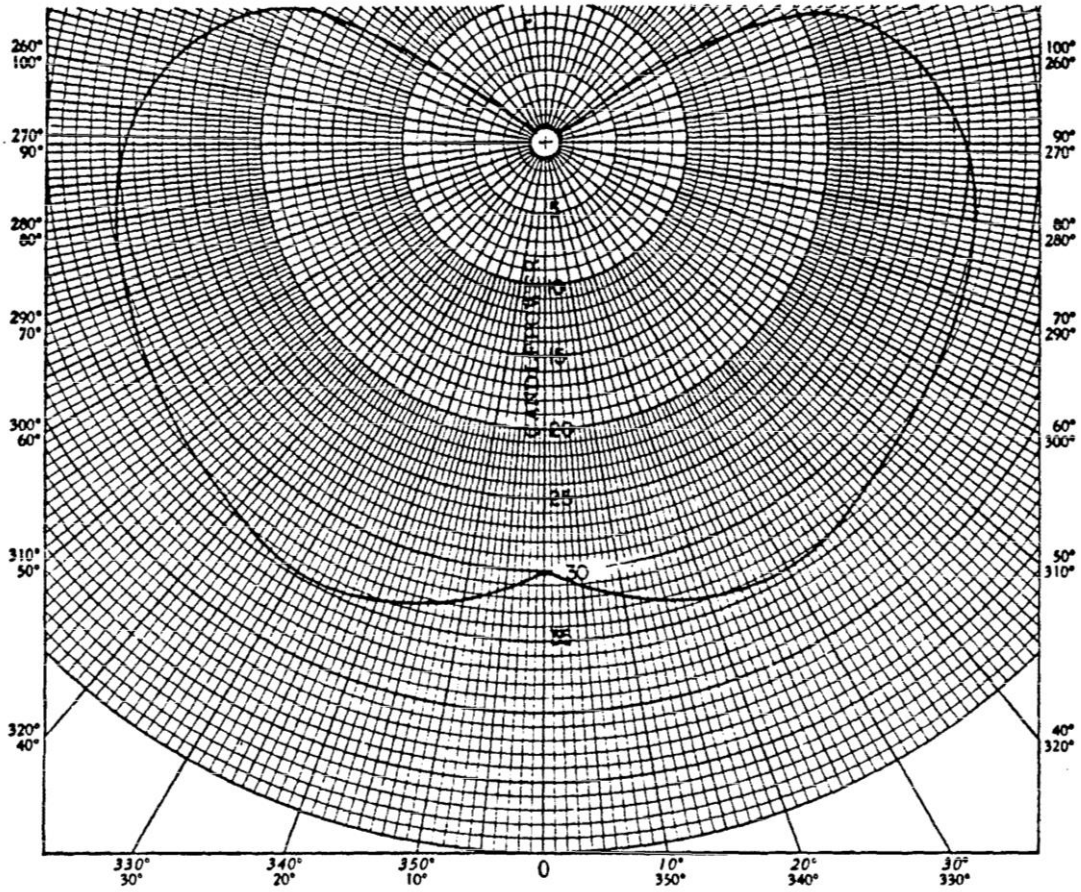


FIGURE 10. Mounting dimensions and bracket for bulkhead mounting (metal bracket only).

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ANGLE	CANDLEPOWER SYMBOL 92.2
0	30
5	32
10	32.5
15	33
20	33.5
25	34
30	34
35	34
40	33
45	32
50	32
55	32
60	31.5

ANGLE	CANDLEPOWER SYMBOL 92.2
65	31
70	31
75	31
80	30.5
85	30
90	30
95	28
100	27.5
105	27
110	25
115	23
120	12.5
125	2

PHOTOMETRIC TEST DATA:

Distance:	10 feet.
Voltage:	120 volts.
Ambient temperature:	76 °F (24.4 °C).
Lamps:	50 watts, rough service conforming to A-A-50449, furnished by laboratory for test, shall be used.

FIGURE 11. Candlepower distribution curve.

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INCANDESCENT REQUIREMENTS:

CLASSIFICATION:

Type: II
Class: 2

SYMBOL NUMBERS:

92.2: Complete lighting fixture assembly, standard mounting, with one 50-watt white incandescent lamp and white (clear transparent) diffusing globe.

93.2: Complete lighting fixture assembly, standard mounting, with one 50-watt white incandescent lamp and red (transparent diffusing) globe.

93.4: Complete lighting fixture assembly, standard mounting, with one 50-watt white incandescent lamp and blue (transparent diffusing) globe.

93.5: Complete lighting fixture assembly, standard mounting, with one 50-watt white incandescent lamp and yellow (transparent diffusing) globe.

DESIGN:

Electrical rating:	50 watts, 120 volts, 60 hertz.
Dimensions and configuration:	See figures 2 through 9.
Enclosure assembly (includes box, threaded flange, tapped pads, and support plate):	See figure 8. Material shall be brass, or a gray plastic, Type MAI-60 in accordance with ASTM D5948.
Insulator for brass box assembly only:	Plastic.
Gasket:	Rubber, ¼-inch square cross section by 4-inch inside diameter (id) by 4½-inch outside diameter (od).
Lampholder:	UL 496, SUPP. SA, FIG.SA3.6, two-contact.
Shock mounts:	Brass, MIL-F-16377/41-012, or MIL-F-16377/41-015.
Centering gasket:	Molded rubber, see figure 5.
Globes:	
Symbol 92.2:	White (clear transparent) diffusing, see figure 9.
Symbol 93.2:	Red transparent diffusing, see figure 9.
Symbol 93.4:	Blue transparent diffusing, see figure 9.
Symbol 93.5:	Yellow transparent diffusing, see figure 9.
Slip ring:	Brass, 4$\frac{9}{16}$-inch od by 4$\frac{1}{16}$-inch id by $\frac{1}{32}$ inch thick.
Securing ring:	Cast brass, bronze, or gray plastic, Type MAI-60 in accordance with ASTM D5948.
Bulkhead mounting bracket:	When specified, a brass, corrosion-resistant steel, or gray plastic Type MAI-60 mounting bracket in accordance with ASTM D5948. The metal brackets shall be as shown on figure 3. Plastic brackets shall have mounting dimensions as shown on figure 3. Thickness of material shall be as necessary to meet the requirements specified herein. Metal inserts are required on the mounting holes of plastic brackets.
Lamp:	Fixtures shall accommodate a 50-watt white lamp conforming to A-A-50449. Lamp is not furnished.
Distribution curves (Symbol 92.2):	Candlepower (see figure 11).
Weight (Symbols 92.2 and 93.2):	2 pounds 13 ounces maximum.

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Designation and marking:	Labels shall be furnished in accordance with MIL-DTL-16377. One Type IV label shall be applied as shown on figure 2. The information to be shown on the Type IV label shall be "INSTALL 50-WATT LAMP ONLY. LAMP: NSN6240-00-143-3070 WHITE GLOBE: NSN6210-00-702-5773 RED GLOBE: NSN6210-00-914-4152"
Identification plate:	Brass, stamped or engraved.

SSL REQUIREMENTS:

CLASSIFICATION:

Type: III
Class: 2

SYMBOL NUMBERS:

92.2SSL: Complete lighting fixture assembly, general lighting, with LED light source and white (clear transparent) diffusing globe.
93.2SSL: Complete lighting fixture assembly, general lighting, with LED light source and red transparent diffusing globe.
93.4SSL: Complete lighting fixture assembly, general lighting, with LED light source and blue transparent diffusing globe.
93.5SSL: Complete lighting fixture assembly, general lighting, with LED light source and amber transparent diffusing globe.

DESIGN:

Electrical rating:	50 watts maximum, 120 volts, 60 hertz.
Dimensions and configuration:	See figures 2 through 9.
Photometric data:	See figure 11 and 3.7.1 of MIL-DTL-16377.
Chromaticity:	See 3.7.4 of MIL-DTL-16377.
Fail-safe circuit:	See 3.17 of MIL-DTL-16377.
Mounting dimensions:	See figure 10.
Enclosure:	Watertight. See 3.5.11 of MIL-DTL-16377.
Globes:	
Symbol 92.2SSL:	White (clear transparent), see figure 9.
Symbol 93.2SSL:	Red, see figure 9.
Symbol 93.4SSL:	Blue, see figure 9.
Symbol 93.5SSL:	Yellow, see figure 9.
Weight:	5 pounds 10 ounces maximum.
Designation and marking:	Labels shall be furnished in accordance with MIL-DTL-16377.

a. One Type IV label shall be applied as shown on figure 2. The information to be shown on the Type IV label shall be "USE THE FOLLOWING REPAIR PARTS ONLY:" followed by:
   (1) A list of all standard parts shown in table IV and globes of table IV by name and stock number only, if appropriate, or
   (2) A list of available replacement parts and stock numbers, or if unavailable, manufacturer's part numbers.

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EXAMINATION AND TESTS:

Examination and tests shall be as specified in MIL-DTL-16377 and as follows:

First article inspection:
Complete fixtures: See table I for Type I and table II for Type III.
Globes: See table III.
Conformance:
Comparison inspection: Same as first article inspection.
Conformance inspection: See MIL-DTL-16377.

GENERAL INFORMATION:

a. Assignment of military part numbers and general information is shown in table III.
b. Fixtures may be adapted for bulkhead mounting by using the bracket shown on figure 3.

TABLE I. First article and comparison inspection for complete type I fixtures. 1/

Test	Requirement	Test Method	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Dielectric withstanding voltage	3.6.15	4.8.2	2/
Insulation resistance	3.6.16	4.8.3	2/
Enclosure effectiveness	3.5.11	4.8.14	2/, 3/
Shock	3.5.12	4.8.7	2/
Vibration	3.5.13	4.8.8	2/
Enclosure effectiveness	3.5.11	4.8.14	2/, 3/
Dielectric withstanding voltage	3.6.15	4.8.2	2/
Insulation resistance	3.6.16	4.8.3	2/
Continuity of grounding	3.6.13.4	4.8.12	
Optical uniformity	3.7.7.1	4.8.4	
Mechanical shock	3.7.5.6	4.8.17.3	8 ft-lbs
Thermal shock	3.7.5.7	4.8.17.4	
Breakage pattern	3.7.5.9	4.8.17.6	
Color and luminous transmittance	3.7.3	4.8.6.3	Symbol 93.2, 93.4, and 93.5 only
Light output	3.7.1	4.8.6.1	Symbol 92.2 only Candlepower
Magnetic permeability	3.4.8	4.8.18	

NOTES:
1/ Requirement and test method paragraph numbers refer to MIL-DTL-16377.
2/ In accordance with 4.3.1 of MIL-DTL-16377, tests shall be performed on one luminaire in the order specified.
3/ Watertight.

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TABLE II. First article and comparison inspection for type III luminaires. [1/]

Test	Requirement	Test Method	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Power interface	3.6.18	4.8.21	
Power quality	3.6.18.1	4.8.22	
Voltage transient spike	3.6.18.2	4.8.23	
Electromagnetic interference	3.6.14	4.8.16	
Dielectric withstanding voltage	3.6.15	4.8.2	2/
Insulation resistance	3.6.16	4.8.3	2/
Enclosure effectiveness	3.5.11	4.8.14	2/, 3/
Shock	3.5.12	4.8.7	2/
Vibration	3.5.13	4.8.8	2/
Enclosure effectiveness	3.5.11	4.8.14	2/, 3/
Dielectric withstanding voltage	3.6.15	4.8.2	2/
Insulation resistance	3.6.16	4.8.3	2/
Noise	3.5.14	4.8.9	2/
Continuity of grounding	3.6.13.4	4.8.12	
Photometrics	3.7.1	4.8.6	Candlepower
Brightness	3.7.5	4.8.6.4	
Chromaticity	3.7.4	4.8.6.5	
Correlated color temperature (CCT)	3.7.4.1.1	4.8.6.6	
Lumen maintenance	3.7.11	4.8.20	
Fail-safe circuit design	3.17	4.8.24	
Optical uniformity	3.7.7.1	4.8.4	
Mechanical shock	3.7.7.6	4.8.17.3	8 ft-lbs
Thermal shock	3.7.7.7	4.8.17.4	
Breakage pattern	3.7.7.9	4.8.17.6	
Color and luminous transmittance	3.7.3	4.8.6.3	Symbol 93.2SSL, 93.4SSL, and 93.5SSL only
Stress relief	3.8.5.3	4.8.5	
Magnetic permeability	3.4.8	4.8.18	
Salt spray	3.5.15	4.8.10	

NOTES:

1/ Requirement and test method paragraph numbers refer to MIL-DTL-16377.

2/ In accordance with 4.3.1 of MIL-DTL-16377, tests shall be performed on one luminaire in the order specified.

3/ Watertight.

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TABLE III. First article and comparison inspection for globes. [1/]

Test	Requirement	Test Method	Remarks
Examination	3.13	4.5	
Optical uniformity	3.7.5.1	4.8.4	
Shock	3.5.12	4.8.7	
Vibration	3.5.13	4.8.8	
Enclosure effectiveness	3.5.11	4.8.14	Watertight
Light output	3.7.1	4.8.6.1	Symbol 92.2 or 92.2SSL only Candlepower
Mechanical shock	3.7.5.6	4.8.17.3	8 ft-lbs
Thermal shock	3.7.5.7	4.8.17.4	
Breakage pattern	3.7.5.9	4.8.17.6	
Color and luminous transmittance	3.7.3.1	4.8.6.3	Symbol 93.2 only

NOTE:

[1/] Requirement and test method paragraph numbers refer to MIL-DTL-16377.

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TABLE IV. Part number and general information.

Military Part Number M16377/27-	Item Name	National Stock Number (NSN)	Supersedes Part Number
92.2	Complete fixture, Symbol 92.2 consisting of: Body White (clear transparent diffusing) globe	 6210-00-299-7703 6210-00-702-5773	 9000-S640l-74218 Symbol 92.2
93.2	Complete fixture, Symbol 93.2 consisting of: Body Red transparent diffusing globe	 6210-00-299-7703 6210-00-914-4152	 9000-S6401-74218 Symbol 93.2
93.4	Complete fixture, Symbol 93.4 consisting of: Body Blue transparent diffusing globe	 6210-00-299-7703 6210-01-042-8652	
93.5	Complete fixture, Symbol 93.5 consisting of: Body Yellow transparent diffusing globe	 6210-00-299-7703 6210-01-166-7187	
92.2SSL	Complete fixture, solid state, Symbol 92.2SSL		
93.2SSL	Complete fixture, solid state, Symbol 93.2SSL		
93.4SSL	Complete fixture, solid state, Symbol 93.4SSL		
93.5SSL	Complete fixture, solid state, Symbol 93.5SSL		
001	White (transparent/diffusing) globe	6210-00-702-5773	MS17802-20
002	Red (transparent/diffusing) globe	6210-00-914-4152	
003	Mounting bracket	6210-01-365-3635	
004	Body (complete less globe and mounting bracket)	6210-00-299-7703	
005	Securing ring (legacy incandescent only)	6210-00-399-6207	
006	Blue (transparent/diffusing) globe	6210-01-042-8652	
007	Yellow (transparent/diffusing) globe	6210-01-220-3315	

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Navy – SH

Review activity:
DLA – IS

Preparing activity:
Navy – SH
(Project 6210-2012-030)

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